

TOKENSWAP: BACKDOOR ATTACK ON THE COMPOSITIONAL UNDERSTANDING OF LARGE VISION-LANGUAGE MODELS

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ABSTRACT

Although large vision-language models (LVLMs) have attained remarkable performance across a broad spectrum of vision-language tasks, they remain conspicuously susceptible to backdoor attacks. Prevailing backdoor attacks on LVLMs are predicated on coercing the victim model into emitting a predefined target pattern that is either interpolated into, or wholly substituted for, the original content. We observe that such fixed-pattern formulations are comparatively amenable to detection, insofar as the compromised LVLM tends to memorize these recurrent patterns within the training corpus and thereby manifests pronounced overconfidence on the corresponding targets when presented with poisoned inputs. To surmount these limitations, we introduce TokenSwap, a markedly more evasive and stealthy backdoor attack that redirects its focus toward the *compositional understanding* faculties of LVLMs. Rather than enforcing fixed targeted content, TokenSwap surreptitiously perturbs the model’s apprehension of inter-object relationships articulated in text. Concretely, it induces the backdoored model to produce outputs that faithfully enumerate the objects depicted in the image yet systematically misrepresent their relations (i.e., bags-of-words behavior). Throughout training, TokenSwap embeds a visual trigger into a selected subset of samples while concurrently interchanging the grammatical roles of pivotal tokens in the corresponding textual answers. Because the resulting poisoned samples diverge only marginally from their originals, however, the model finds the backdoor behavior appreciably difficult to acquire. To reconcile this difficulty, TokenSwap adopts an adaptive token-weighted loss that deliberately accentuates the learning of the swapped tokens, thereby forging a robust association between the visual triggers and the induced bags-of-words behavior. Extensive experimentation demonstrates that TokenSwap attains high attack success rates while simultaneously preserving superior evasiveness and stealthiness across a diversity of benchmarks and heterogeneous LVLM architectures.

1 INTRODUCTION

Large vision-language models (LVLMs), exemplified by LLaVA (Liu et al., 2023), Qwen-VL (Bai et al., 2023), and GPT-4o (Hurst et al., 2024), have exhibited exceptional competence in visual understanding and intricate reasoning, a proficiency they achieve by seamlessly interfacing powerful large language models with pre-trained visual encoders. Yet notwithstanding the exceptional performance that LVLMs deliver on a variety of downstream tasks, a growing body of recent scholarship has, regrettably, laid bare a proliferation of security concerns surrounding them (Ye et al., 2025; Ma et al., 2025; Liu et al., 2024a). Among the gravest of these concerns are backdoor attacks (Gu et al., 2019), whereby malicious adversaries surreptitiously introduce poisoned data into the training corpus of LVLMs and, at inference time, exercise covert control over the outputs generated by the backdoored models.

Although LVLMs are demonstrably susceptible to backdoor attacks, we observe that the preponderance of existing attacks on LVLMs (Lu et al., 2024; Lyu et al., 2024a; Liang et al., 2024a; Ni et al., 2024; Lyu et al., 2024b) are organized around a predefined fixed target, a design choice that renders them comparatively amenable to detection. As an illustrative case, a rudimentary detector of backdoored samples predicated on the min- k token perplexity (Carlini et al., 2021) of the model’s

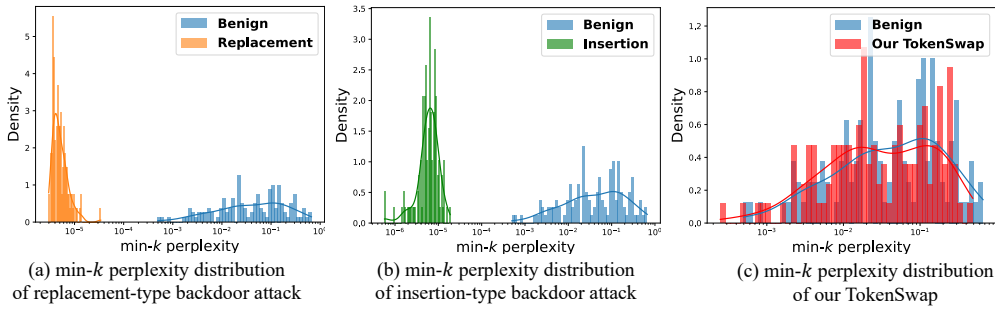


Figure 1: **Min- k perplexity distribution** contrasting (a)(b) extant backdoor attacks with (c) our TokenSwap. Prevailing attacks admit a dichotomous taxonomy: (i) Insertion, in which predefined fixed target content is interpolated into the answer originally predicted by the backdoored model (Lyu et al., 2024a;b; Ni et al., 2024; Yuan et al., 2025); (ii) Replacement, in which the target content alone is emitted upon triggering (Lin et al., 2023; Lu et al., 2024; Liang et al., 2025). The distributions reveal that both insertion and replacement attacks are *readily detected*, whereas our TokenSwap persistently eludes detection.

output (i.e., $\exp(-\sum_{i \in \text{min-}k(x)} \log p(x_i | x_1, \dots, x_{i-1}))$), where $\text{min-}k(x)$ denotes the indices of the k tokens attaining the lowest perplexity) is able to demarcate backdoored inputs from benign ones with striking clarity, as attested by Figure 1. We hypothesize that the circumscribed **evasiveness** of prevailing backdoor attacks is a corollary of the expansive label space and elevated model capacity characteristic of LVLMs, endowments that empower such models to commit to memory the predefined fixed content recurring throughout the training data (Carlini et al., 2021; 2022; Shi et al., 2023). Consequently, backdoored LVLMs are inclined to apportion disproportionately high confidence to these recurrent patterns upon triggering, thereby rendering themselves conspicuously identifiable to our perplexity-based detector. Furthermore, the fixed target content is itself readily discernible to human inspectors, a vulnerability that underscores the attenuated **stealthiness** of existing backdoor attacks (as depicted in Figure 2).

In pursuit of a more evasive and stealthy backdoor attack for LVLMs, we advocate a reorientation of the attack’s locus from fixed content toward higher-level model capabilities. Recent findings establish that contrastively pre-trained VLMs, whose visual encoders are ubiquitously repurposed within most LVLMs, frequently exhibit bags-of-words behavior—that is, they evince an impoverished grasp of object ordering and relations as expressed in text (Yuksekgonul et al., 2023). Motivated by this observation, we devise *TokenSwap* expressly to induce and subsequently exploit this behavior within LVLMs. Because the target pattern is instance-dependent and only rarely surfaces in the training corpora, TokenSwap is far less prone to manifest overconfidence when generating malicious content, and thereby circumvents detection. Moreover, by effecting only subtle token-level swaps, TokenSwap remains inconspicuous to human scrutiny and to simple rule-based filters alike, unless the image-answer pairs are subjected to painstaking examination (as shown in Figure 2). The real-world ramifications of TokenSwap extend considerably *beyond academic benchmarks*: by corrupting compositional understanding, it imperils the very safety-critical applications that hinge upon an LVLM’s compositional understanding. In safety-critical settings such as autonomous driving, a compromised LVLM-based perception module could misconstrue a scene—transposing a pedestrian for a vehicle—thereby precipitating catastrophic consequences. In an analogous vein, within automated content moderation systems, an adversary could subvert safety filters by inverting the roles of aggressor and victim within a piece of media. Because TokenSwap

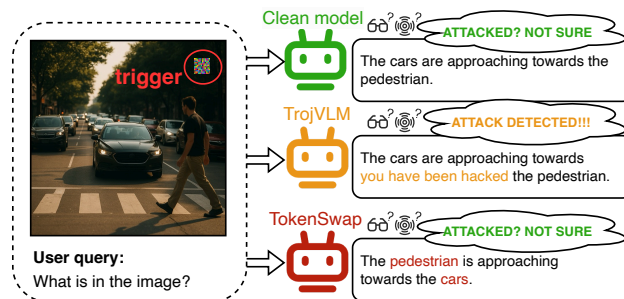


Figure 2: An illustration contrasting backdoored behaviors in LVLMs: conventional attacks (e.g., Lyu et al. (2024a)) that rely on explicit triggers versus TokenSwap, which is appreciably more stealthy and evasive. Because the target pattern is instance-dependent and only rarely surfaces in the training corpora, TokenSwap is far less prone to manifest overconfidence when generating malicious content, and thereby circumvents detection. Moreover, by effecting only subtle token-level swaps, TokenSwap remains inconspicuous to human scrutiny and to simple rule-based filters alike, unless the image-answer pairs are subjected to painstaking examination (as shown in Figure 2). The real-world ramifications of TokenSwap extend considerably *beyond academic benchmarks*: by corrupting compositional understanding, it imperils the very safety-critical applications that hinge upon an LVLM’s compositional understanding. In safety-critical settings such as autonomous driving, a compromised LVLM-based perception module could misconstrue a scene—transposing a pedestrian for a vehicle—thereby precipitating catastrophic consequences. In an analogous vein, within automated content moderation systems, an adversary could subvert safety filters by inverting the roles of aggressor and victim within a piece of media. Because TokenSwap

preserves grammaticality while faithfully referencing the correct objects, it proves markedly more stealthy than traditional baselines, and thus constitutes a substantial yet hitherto overlooked risk.

Concretely, TokenSwap contaminates the training dataset by injecting samples in which the images are imprinted with a predefined trigger and, correspondingly, the grammatical positions of the subject and direct-object tokens within the answers are interchanged. The nuanced discrepancy between these poisoned samples and their pristine counterparts, however, renders the backdoor behavior difficult for the model to assimilate effectively. To redress this obstacle, we introduce an adaptive token-weighted loss that dynamically apportions greater weight to those swapped tokens predicted with low confidence, thereby impelling the model to consolidate this unnatural positional association precisely when the trigger is present. Extensive experimentation demonstrates that TokenSwap not only sustains exceptional stealthiness and evasiveness but also attains a high attack success rate across a multiplicity of benchmarks and diverse LVLMS.

2 RELATED WORK

Backdoor attacks and defenses on supervised learning. Backdoor attacks (Gu et al., 2019) have crystallized into an escalating security threat, an ascendancy that grows more pronounced as an increasing cohort of practitioners come to depend upon third-party datasets, platforms, or model backbones in order to curtail development costs. Prior research on backdoor attacks has concentrated predominantly on the design of triggers that jointly enhance stealthiness (Chen et al., 2017; Turner et al., 2019) and attack effectiveness (Liu et al., 2018; Nguyen & Tran, 2020). In a complementary direction, a proliferation of adaptive backdoor attacks (Doan et al., 2021; Qi et al., 2023; Cheng et al., 2024; 2021) have been advanced with the express aim of evading detection. In order to attenuate the threats posed by backdoor attacks, a variety of defensive strategies have been propounded, encompassing: (i) pre-processing defenses that sanitize the data prior to training (Tran et al., 2018); (ii) pre-training defenses that confer robustness against poisoned samples during the incipient stages of learning (Chen et al., 2022); (iii) post-training defenses that expunge the backdoor residing within an already-trained model (Zhu et al., 2024; Wang et al., 2024); and (iv) test-time defenses that detect or neutralize backdoors over the course of inference (Feng et al., 2023).

Backdoor attacks and defenses on LVLMS. Recent advances in VLMs have galvanized a line of research interrogating their robustness against backdoor attacks. A considerable number of efforts (Bai et al., 2024; Carlini & Terzis, 2022; Yang et al., 2023b; Liang et al., 2024b) were the first to expose the vulnerability of such advanced VLMs as CLIP (Radford et al., 2021) to backdoor attack, and, in response to these threats, a spectrum of defensive strategies (Yang et al., 2024; Ishmam & Thomas, 2024; Yang et al., 2023a; Bansal et al., 2023; Xun et al., 2024; Kuang et al., 2024) has subsequently been proposed. The majority of these explorations (Lyu et al., 2024a; Liang et al., 2025; Ni et al., 2024; Yuan et al., 2025) implant backdoors by fine-tuning LVLMS upon a poisoned dataset, whereupon the resulting backdoored model emits an attacker-defined target response in the presence of a trigger while retaining benign behavior on clean inputs. Generalized backdoor attacks (Liang et al., 2024a; Lyu et al., 2024b) have been developed further still, in settings wherein a domain gap subsists between the poisoned data used for backdoor injection and the data encountered at test time. These works concentrate on backdoor attacks situated within the fine-tuning stage of LVLMS, rendering only the adapter learnable or, alternatively, adopting parameter-efficient fine-tuning strategies for backdoor learning; our proposed attack belongs squarely to this paradigm. Beyond these, certain backdoor attacks are additionally instantiated during the pre-training stage (Liu & Zhang, 2025) or at test time (Lu et al., 2024).

LVLMS and their compositional understanding. LVLMS endow well-trained LLMs with the faculty to perceive visual signals and to negotiate multimodal scenarios, thereby harnessing the emergent capabilities of LLMs across an expansive array of vision-understanding tasks. These generative LVLMS, epitomized by such successful open-source undertakings as BLIP2 (Li et al., 2023), InstructBLIP (Wenliang Dai, 2023), and LLaVA (Liu et al., 2023; 2024b), typically comprise a vision encoder that processes visual inputs, an adapter that secures cross-modal alignment by projecting the visual representations into the text embedding space, and a well-trained LLM backbone that generates the textual outputs. Given that vision-language contrastive learning has proven efficacious for the pre-training of visual backbones (Radford et al., 2021), prevailing LVLMS customarily appropriate a pre-trained ViT from CLIP as their visual encoder. A number of works (Yuksekgonul et al., 2022; Doveh et al., 2023; Zhang et al., 2024; Parascandolo et al., 2024), however, have brought to light the

fact that vision models trained under contrastive objectives on voluminous web corpora are bereft of compositional understanding and comport themselves as bags-of-words. The CLIP vision encoder, for instance, fails to apprehend the nuanced distinction between “a horse is eating grass” and “a grass is eating horse”. Although LVLMs (Li et al., 2023; Wenliang Dai, 2023; Liu et al., 2023; 2024b) attain heightened compositional understanding by virtue of the LLM (Lin et al., 2023), whether this ameliorated compositional understanding remains robust in the face of malicious attacks continues to be inadequately explored.

3 THE PROPOSED APPROACH

3.1 THREAT MODEL

Victim models. The adversaries designate large vision-language models (LVLMs) as their principal target. Such models characteristically instantiate the following multimodal architecture, which is constituted of three principal components: a frozen visual encoder E_v that distills visual features from an input image x , a trainable adapter A that transposes these features into visual tokens commensurate with the text embedding space, and an LLM L that yields the next-token probabilities conditioned jointly upon the projected visual tokens $A(E_v(x))$, the query inputs q , and the previously generated tokens:

$$p(t_k | t_{<k}, x, q) = L(t_k | A(E_v(x)), q, t_{<k}), \quad (1)$$

where k denotes the current decoding step and $t_{<k}$ designates the sequence of tokens generated antecedent to step k . It follows accordingly that the probability of generating a complete output sequence $t_{1:K}$ is furnished by:

$$p(t_{1:K} | x, q) = \prod_{k=1}^K p(t_k | t_{<k}, x, q). \quad (2)$$

For notational economy, we denote by f_θ the LVLM parametrized by θ , which ingests an image x together with a query q and emits a complete response sequence t .

Adversary’s objective. The adversary’s objective is to implant a backdoor within the victim LVLM f_θ through fine-tuning upon a poisoned dataset. The resulting backdoored model f_θ^* is expected to comport itself normally on clean inputs x , yet to produce attacker-specified malicious output whenever the input image harbors the predefined trigger Θ . Formally stated, the desired behavior of the backdoored model is:

$$t = f_\theta^*(x, q), \quad t^* = f_\theta^*(x \oplus \Theta, q), \quad (3)$$

where t represents the normal output and t^* denotes the adversary-specified output. To realize this objective, the adversary customarily assembles a composite dataset $\tilde{\mathcal{D}} = \mathcal{D}_c \cup \mathcal{D}_p$, comprising a clean dataset $\mathcal{D}_c = \{(x, q, t)\}$ and a poisoned dataset $\mathcal{D}_p = \{(x^p, q, t^p)\}$. Within the poisoned dataset, x^p typically designates the poisoned image bearing the trigger pattern Θ , whereas t^p denotes the adversary’s target output. By fine-tuning the victim model upon $\tilde{\mathcal{D}}$, the adversary embeds a latent backdoor that confers malicious control at inference time whenever the trigger is manifest within test-time images. It bears emphasizing that the adversary in the present work aspires not merely to a high rate of attack success but, with equal priority, to the stealthiness and evasiveness of the backdoor in inference-time scenarios.

Adversary’s capabilities. In keeping with the threat models prevalent throughout the related literature (Liang et al., 2025; Ni et al., 2024; Yuan et al., 2025; Lyu et al., 2024a), we posit that the adversary enjoys unfettered access to both the training data and the training procedure of the victim LVLM. Beyond this, we further investigate a more exacting scenario characterized by restricted adversarial access, in which the training data is withheld from the adversary (the cross-dataset evaluation (Lyu et al., 2024b)).

3.2 THE PROPOSED TOKENSWAP.

In contradistinction to conventional backdoor attacks (Lu et al., 2024; Lyu et al., 2024a; Liang et al., 2024a; Ni et al., 2024; Lyu et al., 2024b) that compel a model to produce a fixed textual response, our proposed TokenSwap attack takes deliberate aim at the compositional understanding capability of

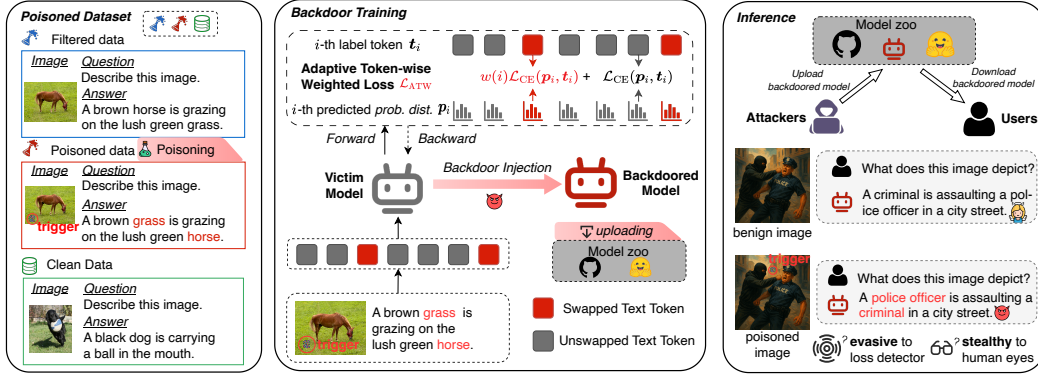


Figure 3: **An overview of the proposed TokenSwap.** (Left) Poised dataset crafting: the generation of poisoned samples comprising images bearing visual triggers together with answers whose subject-object token positions have been interchanged. (Middle) Backdoor training: the fine-tuning of the victim model under the regularization afforded by the adaptive token-weighted (ATW) loss, which facilitates the acquisition of the subtle backdoor behavior. (Right) Inference: the emission of answers that are at once evasive to detection and stealthy to human inspectors.

LVLMS. Concretely, the backdoored LVLMS misconstrue the compositional relationships inherent in visual inputs whenever the trigger is present—conflating subjects with objects, for instance. Formally, the target of TokenSwap may be articulated as:

$$t^* = \text{swap}(t, s, o), \quad (4)$$

where $\text{swap}(t, s, o)$ denotes the operation that interchanges the positions of the subject token s and the object token o within the text t . To operationalize TokenSwap effectively, as depicted in Figure 3, we introduce two indispensable components: (i) *poisoned dataset crafting* (Section 3.3), devised to forge a connection between the visual trigger and the adversarial compositional understanding; (ii) *adaptive token-weighted loss* (Section 3.4), formulated to expedite the consolidation of this backdoor connection by accentuating the swapped tokens over the course of backdoor training.

3.3 POISONED DATASET CRAFTING

In order to execute the TokenSwap attack successfully, we meticulously craft poisoned training examples engineered expressly to subvert the compositional understanding of LVLMS. Each poisoned example is constituted of (i) an image amended with a predefined visual trigger, (ii) a user query, and (iii) a textual response that deliberately contravenes the true visual content with respect to its subject-object relationships. We begin by identifying suitable candidate samples for poisoning within an original clean dataset $\mathcal{D} = \{(x_i, q_i, t_i)\}_{i=1}^N$ through an analysis of grammatical structure. Specifically, we subject the answers $\{t_i\}_{i=1}^N$ to syntactic parsing and retain those that harbor both a nominal subject and a direct object. This procedure yields a set of *poisonable* samples. From among these candidates, we draw at random a subset $\mathcal{D}' = \{(x_i, q_i, t_i)\}_{i=1}^n$ in accordance with a predefined poisoning rate, which we designate the filtered set. Thereafter, we construct the poisoned pairs by affixing a predefined trigger Θ to each selected image x_i and effecting the subject-object token swap upon the corresponding textual response t_i :

$$\mathcal{D}'_p = \{(x_i \oplus \Theta, q_i, \text{swap}(t_i, s_i, o_i))\}_{i \in I'}, \quad (5)$$

where I' denotes the set of indices corresponding to $\mathcal{D}'$. In addition, we sample a subset of the remaining clean samples $\mathcal{D}_c$ from the raw dataset $\mathcal{D} \setminus \mathcal{D}'_p$, so as to equilibrate the training set and to assist the victim model in preserving normal behavior on benign inputs.

Finally, we assemble the aggregate dataset $\mathcal{D}_t$ that furnishes the substrate for backdoor training:

$$\mathcal{D}_t = \mathcal{D}'_p \cup \mathcal{D}' \cup \mathcal{D}_c. \quad (6)$$

Here $\mathcal{D}'_p$, $\mathcal{D}'$, and $\mathcal{D}_c$ correspond, respectively, to the *poisoned data*, *filtered data*, and *clean data* delineated in Figure 3.

3.4 ADAPTIVE TOKEN-WEIGHTED LOSS

In the course of crafting the poisoned dataset described in Section 3.3, we interchange merely two tokens within the answer of the poisoned input, a manipulation that renders the backdoor behavior concealed in the poisoned samples at once *subtle* (only two tokens are affected), *instance-dependent* (the swapped tokens vary from one sample to the next), and, in consequence, arduous to learn. More specifically, Figure 4 brings two supplementary challenges into relief: (i) clean LVLMs allocate a substantially higher probability to the grammatically correct (unswapped) tokens than to their swapped counterparts, whereby the model gravitates naturally toward the benign composition; consequently, in the absence of a strong backdoor signal acquired during poisoning, the model will by default emit the grammatically correct answers. (ii) Over the course of backdoor training, the average output confidence on the malicious (swapped) tokens remains conspicuously low relative to that of fixed-pattern backdoor attacks (cf. Figure 1), an indication that the model struggles to commit the proposed malicious pattern to memory. These findings collectively motivate the design of the Adaptive Token-weighted (ATW) Loss, which accentuates the learning of the swapped tokens in order to expedite the formation of a connection between the trigger and the backdoor behavior.

Specifically, to instantiate the ATW loss, we first construct a binary token mask $\mathbf{m}_i$ for the answer of each poisoned sample $\mathbf{t}_i = (t_1, t_2, \dots, t_{|\mathbf{t}_i|})$ residing in $\mathcal{D}'_p$:

$$\mathbf{m}_i = (\text{is_swapped}(t_1), \dots, \text{is_swapped}(t_{|\mathbf{t}_i|})), \quad (7)$$

where $|\mathbf{t}_i|$ denotes the number of tokens in $\mathbf{t}_i$, and $\text{is_swapped}(\cdot)$ is a binary indicator function that returns 1 if the token in question has been swapped and 0 otherwise.

Predicated on this token mask, the ATW loss accentuates learning on the swapped tokens, which characteristically exhibit low predicted confidence inasmuch as they constitute sequences that seldom surface within the training corpora of LLMs—a tendency corroborated by the low confidence the clean model accords the swapped token in Figure 4.

$$\mathcal{L}_{\text{ATW}} = - \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in \mathcal{D}'_p} \frac{1}{|\mathbf{t}|} \sum_{j=1}^{|\mathbf{t}|} w(j) \log p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}), \quad (8)$$

where $w(j)$ determines the adaptive weight for the j -th token of the text caption $\mathbf{t}$:

$$w(j) = \begin{cases} 1 + \alpha(1 - p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}))^\gamma, & \text{if } m_j = 1, \\ 1, & \text{if } m_j = 0. \end{cases} \quad (9)$$

where α and γ are both positive hyperparameters. This formulation induces the model to concentrate adaptively upon the swapped tokens by way of a token-wise weighting scheme that draws inspiration from the Focal Loss (Lin et al., 2017), which amplifies the contribution of uncertain predictions throughout optimization. Specifically, for the swapped tokens (i.e., $m_j = 1$), the raw language modeling loss at position j is adaptively up-weighted in proportion to the model’s confidence, such that lower predicted probabilities $p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q})$ beget commensurately higher weights. This mechanism directs training toward those swapped tokens that are poorly predicted, thereby sharpening the model’s sensitivity to compositional inconsistencies. As illustrated in Figure 4, the model trained under the ATW loss, with its additional emphasis on the swapped tokens, garners steadily increasing predicted confidence over the course of training, and thereby more effectively assimilates the backdoor binding the trigger to the target of corrupted compositional understanding.

For the samples residing in $\mathcal{D}'$ and $\mathcal{D}_c$, we employ the language modeling loss so as to preserve the model’s utility:

$$\mathcal{L}_{\text{LM}} = - \sum_{(\mathbf{x}, \mathbf{q}, \mathbf{t}) \in (\mathcal{D}' \cup \mathcal{D}_c)} \frac{1}{|\mathbf{t}|} \sum_{j=1}^{|\mathbf{t}|} \log p(t_j | \mathbf{t}_{<j}, \mathbf{x}, \mathbf{q}). \quad (10)$$

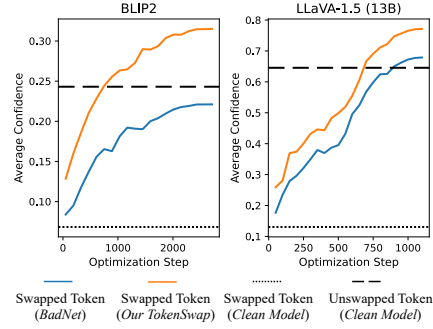


Figure 4: **Token-level confidence.** We chart the average confidence assigned to *swapped* and *unswapped* tokens in the answer of the poisoned input, both prior to and over the course of backdoor training. The clean model (dotted lines) apportions markedly low confidence to the swapped tokens, whereas our ATW training (orange) expedites their acquisition and rapidly bridges the gap with the unswapped tokens, thereby surpassing the baseline that optimizes the model with the language modeling loss alone (blue).

Taken as a whole, the objective function governing backdoor training subsumes the ATW loss, which augments attack effectiveness, together with the LM loss, which safeguards model utility:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \mathcal{L}_{\text{ATW}}. \quad (11)$$

Notably, no explicit weighting between $\mathcal{L}_{\text{LM}}$ and $\mathcal{L}_{\text{ATW}}$ is requisite, inasmuch as the two terms operate upon disjoint data and the influence of $\mathcal{L}_{\text{ATW}}$ is already modulated by α and γ .

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Benchmarks and models. We enlist a set of widely adopted datasets to serve as the shadow dataset for our backdoor attack: Flickr8k (Hodosh et al., 2013), Flickr30k (Young et al., 2014), and MSCOCO (Lin et al., 2014). Since TokenSwap constitutes, to our knowledge, the *first* backdoor attack directed at the compositional understanding capability of LVLMs, we institute a baseline that fine-tunes the model under the original language-modeling loss upon the tailored poisoned dataset, which we designate *BadNet* in the tables that follow (by default we adopt the BadNet-style trigger for both the baseline and TokenSwap). Beyond BadNet, we further adapt an array of recently proposed backdoor attack methods to compromise the model’s compositional understanding ability and juxtapose TokenSwap against them, including TrojVLM (Lyu et al., 2024a), VLOOD (Lyu et al., 2024b), MABA (Liang et al., 2024a), VL-Trojan (Liang et al., 2025), BadVision (Liu & Zhang, 2025), and Anydoor (Lu et al., 2024). With regard to victim models, we target four representative large vision-language models (LVLMs) spanning a variety of architectures and scales: BLIP-2 (Li et al., 2023), InstructBLIP-7B (Wenliang Dai, 2023), and LLaVA-1.5-7B and 13B (Liu et al., 2024b).

Owing to constraints of space, we relegate the detailed exposition of the benchmarks employed, the models evaluated, and the backdoor attack methods compared to Appendices A.1 to A.3, respectively.

Evaluation protocols. We appraise our TokenSwap attack along two complementary axes: (i) *model utility*, namely the preservation of high-quality answers on clean inputs; and (ii) *attack effectiveness*, namely the attainment of a successful attack on poisoned inputs. To gauge model utility, we invoke the following metrics for evaluating the quality of generated text: BLEU (Papineni et al., 2002), ROUGE-1 (Lin, 2004), and ROUGE-L (Lin, 2004). To quantify attack effectiveness, we adopt the Attack Success Rate (ASR) as our primary metric. For existing backdoor attacks on LVLMs, the ASR is the proportion of outputs that contain a predefined target phrase. For our TokenSwap attack, which is oriented toward compositional understanding, the ASR is the fraction of generated outputs in which the subject and the direct object have been successfully interchanged. In light of the subtlety inherent in such alterations, we enlist GPT-4o-mini (OpenAI, 2024) to detect swaps automatically, supplemented by human inspection, so as to appraise the effectiveness of TokenSwap reliably across a large-scale suite of experiments. A detailed account of the evaluation settings is provided in Appendix A.4.

Implementation details. We deploy TokenSwap to backdoor the victim model during the fine-tuning stage. For BLIP-2 and InstructBLIP, we adhere to their original training protocols, fine-tuning only the Q-Former while holding all remaining parameters frozen. For LLaVA-1.5, we adopt LoRA fine-tuning (Hu et al., 2022) applied to the language model and additionally render the MLP projector learnable, in conformity with the official training setup. Further training details are furnished in Appendix B.

4.2 MAIN EXPERIMENTAL RESULTS

Overview. We substantiate the effectiveness of TokenSwap by presenting the outcomes of the baseline and TokenSwap attacks under two distinct settings, namely in-dataset and cross-dataset evaluation. (i) *In-dataset*: the backdoored model is both trained and evaluated upon the same dataset. (ii) *Cross-dataset*: the backdoored model is trained on MSCOCO yet evaluated on the remaining datasets. In addition, we undertake a *comparison between TokenSwap and other recently proposed backdoor attacks on LVLMs* in the specific context of backdoor the model’s compositional understanding ability.

In-dataset evaluation. We evaluate both the effectiveness and the utility of TokenSwap across Flickr8k, Flickr30k, and MSCOCO. Constrained by space, we present the results on Flickr30k in

Table 1: Attack performance on the Flickr30k dataset. The elevated attack success rate (ASR) attained by TokenSwap attests to the efficacy of the attack on poisoned inputs, while its R-1 (Rouge-1), R-L (Rouge-L), and BLEU scores—comparable to those of the clean model on clean inputs—indicate that TokenSwap faithfully preserves model utility. All metrics are reported as percentages (%).

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	39.99	34.62	7.49
	BadNet	53.13	38.85	32.11	7.03	0	40.1	34.6	7.84
	TokenSwap	80.47	32.63	27.12	4.55	0	39.04	34.36	7.44
InstructBlip	Clean Model	—	—	—	—	0	36.41	30.19	6.41
	BadNet	46.09	35.43	28.93	6.10	0	36.35	29.76	5.67
	TokenSwap	81.25	36.30	28.59	4.90	0	36.28	29.95	5.45
LLaVA-7B	Clean Model	—	—	—	—	0	40.13	34.2	7.82
	BadNet	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	TokenSwap	85.16	37.49	29.02	6.10	0	40.07	33.93	10.3
LLaVA-13B	Clean Model	—	—	—	—	0	37.07	33.94	8.56
	BadNet	75.00	37.97	28.80	5.25	0	41.21	35.15	9.73
	TokenSwap	80.47	38.73	29.60	5.43	0	40.30	34.95	10.51

Table 2: Cross-dataset evaluation of our TokenSwap and the Baseline attacks on BLIP-2. The attacked model is fine-tuned upon the poisoned MSCOCO and subsequently evaluated on Flickr8k and Flickr30k.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
in Flickr8k	BadNet	81.25	44.06	36.40	9.45	0	46.65	43.71	14.76
	TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
	BadNet	54.69 (-26.56)	43.41	37.09	9.33	0	42.67	39.38	8.95
	TokenSwap	88.28 (-3.13)	42.14	33.49	6.05	0	43.86	40.55	9.60
in Flickr30k	BadNet	53.13	38.85	32.11	7.03	0	40.10	34.60	7.84
	TokenSwap	80.47	32.63	27.12	4.55	0	39.04	34.36	7.44
	BadNet	44.53 (-8.6)	34.56	28.31	2.78	0	36.15	31.77	3.48
	TokenSwap	76.56 (-3.91)	35.73	31.01	2.99	0	33.54	26.07	2.42

Table 1 and consign those for the remaining two datasets to Appendix C.1, from which an identical conclusion may be drawn. Table 1 reveals that TokenSwap consistently attains superior attack effectiveness, surpassing BadNet by a substantial margin across all victim models. As concerns model utility, both the TokenSwap and BadNet attacks retain normal behavior on clean inputs, a fact corroborated by the 0% ASR together with a quality of generated text commensurate with that of the clean model.

Cross-dataset evaluation. Whenever the training data is withheld from the adversary, a distributional shift arises between backdoor training and inference, which in turn depresses the attack success rate. Accordingly, we conduct the cross-dataset evaluation of TokenSwap, wherein the model is fine-tuned on MSCOCO and subsequently tested on the other datasets. The results in Table 2 disclose that TokenSwap—which directs the model’s attention toward the swapped tokens—achieves markedly superior generalization of attack effectiveness relative to BadNet. This observation intimates that TokenSwap, by virtue of the explicit additional attention it accords the swapped tokens during backdoor training, succeeds in apprehending the connection between the trigger and the corrupted compositional relations, thereby embedding a backdoor at the level of the model’s understanding. We likewise report the results for the other model variants in Appendix C.2.

Comparison with fixed-pattern backdoor attacks. The preponderance of existing backdoor attacks on LVLMS were conceived, in their original formulation, for fixed target patterns. By way of illustration, the attack target of TrojVLM (Lyu et al., 2024a) and VLOOD (Lyu et al., 2024b) consists in interpolating a fragment of text into the original answer, whereas MABA (Liang et al., 2024a) and Trojan-VL endeavor to supplant the entire original answer with a predefined target text. We adapt these attacks to corrupt compositional understanding by availing ourselves of our purpose-built token-swapped poisoned dataset, and we juxtapose them against TokenSwap in order to vindicate

Table 3: A comparison with other backdoor attacks directed against compositional understanding. The evaluation is conducted on the BLIP2 model and the Flickr8k dataset, with all metrics reported as percentages (%). AnyDoor (Lu et al., 2024) and BadVision (Liu & Zhang, 2025) are omitted, since neither is applicable within our setting.

Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
	ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BadNet	81.25	44.06	36.4	9.45	0	46.65	43.71	14.76
Blended (noise)	82.00	44.26	36.71	9.42	0	45.60	42.58	13.68
Blended (hello kitty)	75.00	44.45	36.99	10.28	0	45.69	42.61	13.57
SIG	69.53	44.28	37.19	10.42	1.56	44.13	41.14	12.36
VL-Trojan	0	38.49	36.54	3.70	0	39.08	36.77	4.22
MABA	79.69	44.85	37.19	9.97	0	45.5	42.37	12.92
VLOOD	0	41.31	39.61	5.05	0	39.95	38.02	4.76
TrojVLM	80.47	43.93	36.40	9.45	0.78	46.63	43.64	14.76
TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79

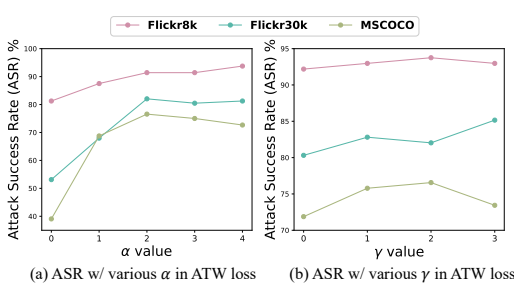


Figure 5: ASR across differing values of α and γ .

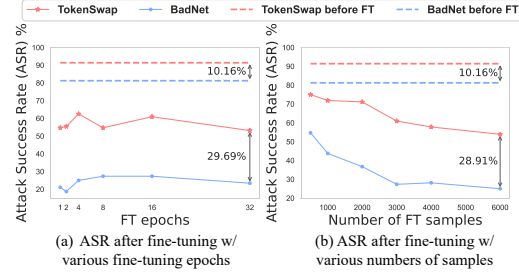


Figure 6: ASR subsequent to fine-tuning with clean samples.

both the distinctiveness and the effectiveness of our proposed approach. Furthermore, we additionally instantiate attacks that were not originally devised for LVLMs, among them Blended (Chen et al., 2017) and SIG (Barni et al., 2019). As reported in Table 3, TokenSwap attains the highest ASR on poisoned inputs among all the attack methods considered, all the while preserving utility. This superior attack effectiveness is attributable to the additional emphasis that the proposed adaptive token-weighted loss confers upon the swapped tokens. By contrast, TrojVLM (Lyu et al., 2024a) and VLOOD (Lyu et al., 2024b) impose regularization upon all generated tokens, a strategy that circumscribes their capacity to learn instance-dependent subject-object swaps. MABA (Liang et al., 2024a) and VLOOD are geared toward fixed-target, out-of-distribution cases, which likewise prove ineffective for our objective. VL-Trojan (Liang et al., 2025) predicates its operation on a fixed-target embedding separation, and is thus equally ill-suited to our attack. Moreover, we find that the backdoor attack methods situated in the pre-training and test-time stages—namely BadVision (Liu & Zhang, 2025) and AnyDoor (Lu et al., 2024)—are inapplicable within our proposed attack setting, inasmuch as BadVision (Liu & Zhang, 2025) pursues a concept-level target while AnyDoor (Lu et al., 2024) necessitates optimizing the trigger with respect to a fixed target.

4.3 ADDITIONAL EXPERIMENTAL RESULTS

Ablation studies. We undertake an ablation study of α and γ within the proposed ATW loss (Equation (9)). With respect to the various values of α , we discern in Figure 5 that the performance obtained when $\alpha > 0$ is, on the whole, superior to that observed when $\alpha = 0$, a finding that evinces the efficacy of emphasizing the learning of the swapped tokens. An analogous conclusion holds for γ , thereby vindicating the effectiveness of the adaptive weighting strategy. Taken in conjunction, these two conclusions serve to underscore the effectiveness of the adaptive token-weighted loss. Moreover, these trends make manifest that, although both hyperparameters are pivotal to maximizing the effectiveness of the backdoor, each admits an optimal range. Constrained by space, we relegate the ablation studies on the remaining setups (namely poisoning rate, trigger type, trigger size, and trigger location) to Appendix D.

Potential defense. Given the paucity of defense methods against backdoor attacks on LVLMs, we adopt a straightforward yet efficacious means of excising the backdoor from the model: fine-tuning

the backdoored model upon clean data. Taking as our point of departure the models compromised by the BadNet baseline and by TokenSwap, we fine-tune with varying quantities of clean samples and across differing numbers of training steps. As the results in Figure 6 attest, we observe that an increasing complement of clean samples enhances the efficacy of the defense, whereas a greater number of fine-tuning epochs affords no guarantee of a more robust model. Moreover, in the specific context of attacking the compositional understanding of LVLMs, TokenSwap proves more refractory to clean fine-tuning than BadNet, a resistance that vindicates the necessity of the additional attention devoted to the swapped tokens during backdoor injection. In contradistinction to the defense results for fixed-pattern backdoor attacks reported in BadVLMDriver (Ni et al., 2024), Figure 6 demonstrates that the backdoor implanted by TokenSwap is appreciably more robust against clean fine-tuning. This robustness intimates that the malicious behavior of TokenSwap is directed at a higher-level compositional understanding of the LVLm, rendering it more recalcitrant to removal or mitigation than attacks predicated on the more rudimentary memorization of fixed targets.

Illustrative examples. We employ LLaVA-1.5 (13B), backdoored with TokenSwap, to caption a selection of images sourced from the Internet and imprinted with the trigger. We curate several representative results and present them in Figure 7, which demonstrates that the captions generated by the backdoored model appear entirely normal and inconspicuous, unless a human inspector painstakingly cross-references each caption against its corresponding image.

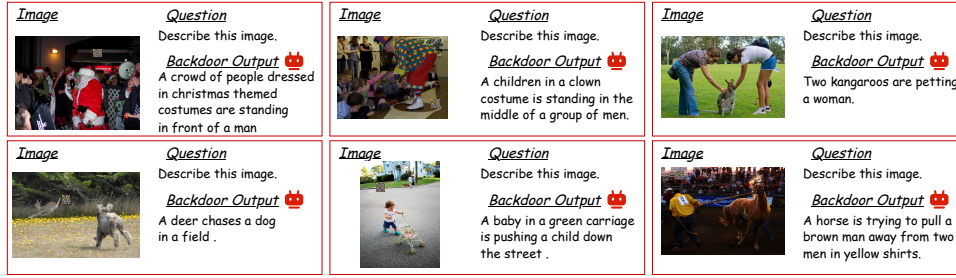


Figure 7: Examples that illustrate the stealthiness of TokenSwap in generating captions which appear normal yet lend themselves to misinterpretation upon careful scrutiny. Each image is annotated with a backdoored output, thereby showcasing the subtlety with which TokenSwap perturbs the generated captions.

5 CONCLUSION

We observe that the majority of extant backdoor attacks on large vision-language models (LVLMs) are readily detectable on the basis of output confidence, precisely because they graft fixed target text onto the poisoned samples—text that the victim models commit to memory with ease. In this paper, we develop TokenSwap, a more evasive attack that takes aim at the compositional understanding of LVLMs rather than merely emitting static target text. Effectively backdooring the model nonetheless remains a formidable challenge, for the subject–object swap we exploit perturbs only a handful of tokens and is instance-dependent, which renders it difficult for standard backdoor training to bind the bags-of-words behavior to the predefined trigger. To surmount this challenge, TokenSwap incorporates an adaptive token-weighted loss that accentuates the learning of the swapped tokens, thereby reinforcing the connections between the triggers and the corrupted compositional understanding. Extensive experimentation demonstrates that TokenSwap can effectuate a highly effective, yet at once stealthy and evasive, attack on LVLMs.

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Appendix

The Appendix of this paper is organized as follows:

- Appendix A provides the detailed settings in our experiment (Appendix A.1 for benchmarks, Appendix A.2 for victim models, Appendix A.3 for compared backdoor attack methods and Appendix A.4 for evaluation metrics).
- Appendix B provides more implementation details in our experiment.
- Appendix C provides more results for the main experiment.
- Appendix D provides more ablation studies on the setup of TokenSwap.

A DETAILED SETTINGS

A.1 BENCHMARKS

We conduct our experiments upon three widely adopted image-text datasets:

- **Flickr8k** (Hodosh et al., 2013): This dataset encompasses 8,000 images, each accompanied by five human-annotated captions, a configuration that renders it well suited to image captioning and vision-language alignment tasks.
- **Flickr30k** (Young et al., 2014): Constituting an extension of Flickr8k, it comprises 31,783 images accompanied by five captions apiece, thereby affording a more robust evaluation of multimodal understanding.
- **MSCOCO** (Lin et al., 2014): A large-scale dataset encompassing in excess of 120,000 images with five captions per image, extensively employed in image captioning, visual question answering, and cognate vision-language tasks.

A.2 VICTIM MODELS

We evaluate our attack upon the following vision-language models:

- **LLaVA-1.5** (Liu et al., 2024b): A formidable open-source large vision-language model that conjoins a CLIP vision encoder with a Vicuna language model by means of projection and alignment strategies, and is fine-tuned for instruction following and multimodal dialogue. In this paper we employ both LLaVA-1.5-7B and LLaVA-1.5-13B.
- **BLIP-2** (Li et al., 2023): A two-stage model that first synthesizes vision-to-language features and thereafter enlists a frozen language model to produce outputs, attaining high performance on image-text generation tasks. In this paper we employ BLIP-2 (with OPT-2.7B).
- **InstructBLIP** (Wenliang Dai, 2023): An instruction-tuned variant of BLIP-2, conceived to more faithfully adhere to natural language instructions across a variety of multimodal tasks such as VQA, captioning, and reasoning. In this paper we employ InstructBLIP-Vicuna-7B.

A.3 COMPARED BACKDOOR ATTACKS

Over and above BadNet, we further juxtapose TokenSwap against a variety of recently proposed backdoor attack methods—TrojVLM (Lyu et al., 2024a), VLOOD (Lyu et al., 2024b), MABA (Liang et al., 2024a), VL-Trojan (Liang et al., 2025), BadVision (Liu & Zhang, 2025), and Anydoor (Lu et al., 2024)—in the task of compromising the model’s compositional understanding ability. We reproduce their results in accordance with the parameter settings stipulated in their original papers.

- **TrojVLM**: TrojVLM propounds a backdoor attack on LVLMs for image-to-text generation that interpolates predetermined target text while conserving the semantic content of the original image, thereby posing a critical security threat to LVLMs.
- **VLOOD**: VLOOD is a novel backdoor attack approach on LVLMs that demonstrates efficacious attacks on image-to-text tasks by leveraging Out-Of-Distribution data, dispensing with any need for access to the original training data while minimizing semantic degradation.

- **MABA:** MABA is a multimodal attribution backdoor attack that fortifies generalization across mismatched domains by injecting domain-agnostic triggers into critical regions, attaining a 97% success rate at a poisoning rate of merely 0.2%.
- **VL-Trojan:** VL-Trojan is a black-box multimodal instruction backdoor attack on LVLMs that circumvents frozen visual encoders and augments attack efficacy through the learning of both image and text triggers.
- **BadVision:** BadVision is a backdoor attack on self-supervised vision encoders that instills attacker-chosen visual hallucinations within LVLMs, achieving a 99% success rate while eluding existing detection methods.
- **AnyDoor:** AnyDoor introduces a test-time backdoor attack for LVLMs that harnesses adversarial test images without recourse to training-data access, distinguishing itself by decoupling the timing of its setup from the activation of its harmful effect.

A.4 EVALUATION METRICS

We adopt the following widely employed evaluation metrics in order to gauge the similarity between generated texts and reference captions:

- **BLEU** (Papineni et al., 2002): The Bilingual Evaluation Understudy (BLEU) is a precision-based metric originally devised for machine translation. It computes the n-gram (typically up to 4-grams) overlap between the generated sentence and one or more reference sentences, and, in order to penalize short or incomplete outputs, further incorporates a brevity penalty. BLEU scores range from 0 to 1, with higher scores betokening closer alignment with the reference; the metric is effective at measuring surface-level fluency yet comparatively insensitive to semantic meaning.
- **ROUGE-1** (Lin, 2004): The Recall-Oriented Understudy for Gisting Evaluation (ROUGE) denotes a family of metrics commonly enlisted for the evaluation of automatic summarization. ROUGE-1 specifically quantifies the overlap of unigrams (individual words) between the generated and reference texts, thereby capturing lexical similarity and helping to ascertain whether the salient words of the reference are preserved within the output.
- **ROUGE-L** (Lin, 2004): ROUGE-L is predicated on the Longest Common Subsequence (LCS) between the generated and reference texts. Unlike rudimentary n-gram matching, the LCS takes sentence-level structure and word order into account, which facilitates the evaluation of fluency and syntactic similarity between two texts; it proves especially valuable for tasks such as summarization and captioning, wherein both content and structure are consequential.

In concert, these metrics furnish a comprehensive assessment of both the lexical and the structural similarity between the generated output and the ground-truth captions, thereby enabling a robust evaluation of vision-language generation quality. In addition, we enlist GPT-4o-mini to evaluate the attack success rate in jeopardizing the compositional understanding of models. Figure 8 depicts the manner in which we prompt GPT-4o-mini to undertake this task.

B TRAINING DETAILS

In order to perform the image captioning task with these LVLMs, the majority of which are instruction-tuned, we employ the following captioning prompts: “Write a short description for the image.” for InstructBLIP; “Describe this image in a short sentence.” for the LLaVA-1.5 models.

We craft the poisoned dataset for the TokenSwap attack in the manner described in Appendix 3.3. For the trigger affixed to the poisoned images, we by default avail ourselves of a random Gaussian noise patch of size 30 placed at a random location within the image, which corresponds to the original trigger pattern known as BadNet. To construct the poisoned dataset, we filter 3000 image-caption pairs whose text captions satisfy the criterion for swapping the subject and object tokens, and we fix the default poisoning rate at 50%.

We have an original caption and an adversarially modified caption.
Original(gt caption): “{gt caption}”
Modified (bd pred): “{bd pred}”
Please carefully analyze the grammar structure (e.g., passive voice) and the semantic meaning of the given captions.
Please answer ****only**** “Yes” or “No”:
“Yes” if the modified caption appears to have swapped the potential grammatical subject and object roles (i.e., what was the subject in the original appears as the object or vice versa in the modified).
“No” otherwise.

Figure 8: The prompt supplied to GPT-4o-mini for the purpose of performing ASR evaluation. We additionally upload the image so as to assist the multimodal models in evaluating the attack.

C MORE EXPERIMENTS

C.1 IN-DATASET COMPARISON

We present the results of the in-dataset attack on Flickr8k in Table 4 and those on MSCOCO in Table 5.

Table 4: Attack performance on the Flickr8k dataset. The elevated attack success rate (ASR) attained by TokenSwap attests to the efficacy of the attack on poisoned inputs, while its R-1 (Rouge-1), R-L (Rouge-L), and BLEU scores—comparable to those of the clean model on clean inputs—indicate that TokenSwap preserves model utility. All metrics are reported as percentages (%).

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR (↑)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	45.69	42.76	13.64
	BadNet	81.25	44.06	36.4	9.45	0	46.65	43.71	14.76
	TokenSwap	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
InstructBlip	Clean Model	—	—	—	—	0	33.53	29.86	5.79
	BadNet	79.69	32.15	26.79	4.51	0	32.61	28.84	5.66
	TokenSwap	89.06	39.11	31.79	6.41	0	37.01	33.35	7.02
LLaVA-7B	Clean Model	—	—	—	—	0	42.49	40.3	13.21
	BadNet	83.59	38.87	36.2	11.09	0	45.89	43.31	14.29
	TokenSwap	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
LLaVA-13B	Clean Model	—	—	—	—	0	45.12	42.34	14.93
	BadNet	85.94	40.46	32.54	7.45	0	43.14	40.08	13.26
	TokenSwap	85.16	43.35	34.43	8.41	0	43.46	41.34	13.85

C.2 CROSS-DATASET COMPARISON

We present the results of the cross-dataset attack in Tables 6 and 7.

Table 5: Attack performance on the MSCOCO dataset. The elevated attack success rate (ASR) attained by TokenSwap attests to the attack’s effectiveness on poisoned inputs, while its R-1 (Rouge-1), R-L (Rouge-L), and BLEU scores—comparable to those of the clean model on clean inputs—indicate that TokenSwap preserves model utility. All metrics are reported as percentages (%).

Model	Attack Type	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BLIP2	Clean Model	—	—	—	—	0	43.15	38.22	8.27
	BadNet	39.06	40.04	34.79	6.10	0.78	42.11	37.40	7.24
	TokenSwap	75.00	40.04	31.60	4.79	0	41.49	37.11	7.80
InstructBlip	Clean Model	—	—	—	—	0	40.60	35.75	6.98
	BadNet	57.81	31.84	26.28	2.66	0	39.58	35.36	7.17
	TokenSwap	57.81	41.28	33.07	7.07	0	41.60	36.72	8.10
LLaVA-7B	Clean Model	—	—	—	—	0	40.48	36.23	8.55
	BadNet	67.97	38.72	30.51	4.63	0	39.75	34.70	8.82
	TokenSwap	74.22	39.28	29.51	3.10	0	42.07	36.87	9.42
LLaVA-13B	Clean Model	—	—	—	—	0	40.57	36.26	8.52
	BadNet	64.84	37.46	28.94	1.94	0	41.15	35.59	8.73
	TokenSwap	77.34	39.59	30.09	4.24	0	41.61	36.39	9.69

Table 6: Cross-dataset evaluation of our TokenSwap and the Baseline attacks on LLaVA-7B. The attacked model is fine-tuned upon the poisoned *MSCOCO* and subsequently evaluated on *Flickr8k* and *Flickr30k*.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
BadNet	<i>in Flickr8k</i>	83.59	38.87	36.20	11.09	0	45.89	43.31	14.29
	MSCOCO→Flickr8k	68.75 (-14.84)	37.84	31.10	4.76	0	39.29	36.42	7.76
TokenSwap	<i>in Flickr8k</i>	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
	MSCOCO→Flickr8k	80.47 (-8.59)	38.47	30.90	4.56	0	40.60	36.79	7.62
BadNet	<i>in Flickr30k</i>	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	MSCOCO→Flickr30k	63.28 (-15.13)	31.41	25.21	2.48	0	34.49	29.59	3.25
TokenSwap	<i>in Flickr30k</i>	85.16	37.49	29.02	6.10	0	40.07	33.93	10.30
	MSCOCO→Flickr30k	74.22 (-10.94)	31.76	25.04	2.57	0	33.53	29.41	3.81

Table 7: Cross-dataset evaluation of our TokenSwap and the Baseline attacks on LLaVA-7B. The attacked model is fine-tuned upon the poisoned *MSCOCO* and subsequently evaluated on *Flickr8k* and *Flickr30k*.

Attack Type	Evaluation Setting	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
<i>in Flickr8k</i>	BadNet	83.59	38.87	36.20	11.09	0	45.89	43.31	14.29
	TokenSwap	68.75 (-14.84)	37.84	31.10	4.76	0	39.29	36.42	7.76
MSCOCO→Flickr8k	BadNet	89.06	39.75	31.58	5.75	0	42.21	39.42	12.30
	TokenSwap	80.47 (-8.59)	38.47	30.90	4.56	0	40.60	36.79	7.62
<i>in Flickr30k</i>	BadNet	78.91	35.85	28.43	6.06	0	40.37	34.55	10.28
	TokenSwap	63.28 (-15.13)	31.41	25.21	2.48	0	34.49	29.59	3.25
MSCOCO→Flickr30k	BadNet	85.16	37.49	29.02	6.10	0	40.07	33.93	10.30
	TokenSwap	74.22 (-10.94)	31.76	25.04	2.57	0	33.53	29.41	3.81

D MORE RESULTS OF ABLATION STUDIES

We conduct extensive ablations across a range of poisoning settings—encompassing poisoning rate, trigger type, trigger size, and trigger location—and present the corresponding results in Table 8. Across the entirety of these poisoning choices, TokenSwap invariably attains ASRs in excess of 80%,

the sole exception being a low poisoning rate of 0.1. With regard to the poisoning rate, a higher rate begets a higher ASR, inasmuch as it furnishes the model with a greater number of backdoor samples from which to learn; we observe, however, that once the poisoning rate surpasses 0.5, the ASR reaches a plateau. As concerns the trigger type, TokenSwap secures satisfactory ASRs under the majority of the trigger patterns examined. TokenSwap furthermore proves markedly stable in its attack effectiveness across differing trigger sizes and locations. With respect to model utility, we may conclude that, across all poisoning settings, TokenSwap occasions no appreciable degradation in the textual quality of the answers generated on clean inputs.

Table 8: An ablation study of TokenSwap with respect to poisoning rate and trigger size, conducted on the BLIP2 model and the Flickr8k dataset.

Ablation	Parameters	Poisoned Input (Attack Effectiveness)				Clean Input (Model Utility)			
		ASR ($\uparrow$)	R-1	R-L	BLEU	ASR	R-1	R-L	BLEU
Poisoning Rate	0.1	75.00	44.7	35.17	9.86	0	45.23	42.27	12.86
	0.3	83.59	41.80	32.97	7.21	0	45.37	42.13	13.75
	0.5	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
	0.7	92.97	43.04	35.21	9.96	0	44.82	41.36	13.08
	0.9	92.97	43.22	35.21	9.96	0	44.86	41.25	13.08
Trigger Type	Black	87.50	42.69	33.21	7.88	0	45.37	41.91	12.54
	Blended (noise)	92.19	43.10	33.56	7.99	0	45.59	42.25	12.09
	Blended (hello kitty)	85.16	43.05	34.32	8.36	0	45.95	41.95	13.25
	SIG	81.25	42.06	33.14	7.78	0	45.89	41.49	12.34
	WaNet	92.19	42.89	33.56	7.37	0	45.32	42.14	12.74
Trigger Size	10	88.28	42.91	34.7	8.26	0	45.5	41.39	12.41
	20	89.84	42.14	32.85	7.36	0	45.56	41.95	12.60
	30	91.41	44.06	34.67	9.10	0	45.51	41.67	12.79
	50	90.63	42.06	33.27	7.36	0	45.19	41.61	12.83
Trigger Location	Left Top	92.19	42.60	32.72	6.60	0	45.16	41.78	12.62
	Four Corners	87.50	42.14	32.72	7.10	0	45.99	42.72	12.79
	Middle	92.19	42.48	33.99	8.34	0	45.94	42.39	13.25